

The Resolution Programme

Obstruction, price, and proof for systems that must act on less than what is there — the grand picture that emerges from unifying eleven research lines: causal descent, bounded transduction, quantum interfaces, cell metabolism, planetary governance, and language agents.

Amin Abaee

Independent Researcher, Tehran · ORCID 0000-0002-0019-1842

Contents

1 · The One-Sentence Picture 2

2 · The Primitive: Factoring Through a Bottleneck 3

3 · One Constraint, Many Disguises 4

4 · The Ladder of Assembly and the Hankel Bridge 6

5 · The Three Faces of Every Result 7

6 · The Two Walls and the Price Schedule 8

7 · The Corpus Map: All Works on One Grid 9

8 · The Quantum Wing 11

9 · The Empirical Landing: Biology as a Working Instance 12

10 · The Governance Scale 13

11 · The AI Wing: Bounded Observers Made of Language 14

12 · The Compositional Home 14

13 · The Methodological Spine 15

14 · The Open Frontier 16

1 • The One-Sentence Picture

Read from above, the corpus is one mathematics. Every line of work — the causal-descent monographs, the rate-distortion theory of bounded sequential transduction, the quantum-comb and instrument papers, the monitored-circuit manuscript, the metabolic-curvature pair, the nine governance papers of the sustainability arena, the chess paper — studies the same object: a system whose behaviour must factor through less information than the world contains. Before asking which behaviour is optimal, the corpus asks which behaviour can be realized at all by an agent with specified observations, causal access, memory, computational resources, and standards of evidence. And when the answer is no, it asks precisely how far short any permitted implementation must fall, and what independently checkable artifact witnesses that answer.

Three questions therefore recur in every work, each more refined than the last. Existence: when do locally acceptable behaviours assemble into a global one that survives the bottleneck? Price: what does it cost, in sensors, memory, width, or bits, to widen the bottleneck — and if you cannot, how much error is unavoidable? Proof: when a request fails, what finite, independently checkable object shows it, and where does the failure live? The corpus answers all three, at scales from a two-intersection driving puzzle to a monitored quantum circuit, from a single ribosome-scale metabolic network to planetary governance.

The master file “top-down of my work.txt” records four independent syntheses of this corpus, produced by four different models, each given full descriptions of only two works: (a) effective causal descent and (b) the rate-distortion transduction manuscript. All four converged, without coordination, on the same reading of those two pillars — a theory of constrained realizability (astra); existence, proof, and price under observational uniformity (opus 5.5); a rate-distortion theory of bounded observers, with certificates (fable); the Resolution Programme (opus 5). All four also flagged the same limitation: the remaining works arrived as bare identifiers and DOIs, unresolvable from the descriptions supplied. This document closes that loop. Every work in the corpus has now been read at line level; the slot-grids the four syntheses left open can be filled; the picture can finally be stated over the whole of it.

One primitive — behaviour factors through what the agent can distinguish. Two currencies — width and distortion. Three faces — existence and obstruction, price and frontier, proof and certified synthesis. Two walls — combinatorial and analytic. One compositional home, and one epistemology: charts carry certificates, ladders carry monotonicity, limits carry invariants.

A serviceable name for the whole remains the one opus 5 gave it: the Resolution Programme — obstruction, price, and proof for systems that must act on less than what is there. The chapters that follow unfold that sentence: the primitive (Chapter 2), its disguises (Chapter 3), the ladder of assembly (Chapter

4), the three faces (Chapter 5), the two walls and the price schedule (Chapter 6), the corpus map with every work docked (Chapter 7), and then the wings — quantum (Chapter 8), biological (Chapter 9), governance (Chapter 10), linguistic (Chapter 11) — before the compositional home (Chapter 12), the epistemology (Chapter 13), and the open frontier (Chapter 14).

2 • The Primitive: Factoring Through a Bottleneck

The formal situation is invariant across the corpus. There is a hidden world or history h , and an observational record $O(h)$ — what the sensors return, what the prefix of the past permits, what the finite state can retain, what the interface transmits. A deterministic implementable policy must factor through that record: $\pi = \pi' \circ O$. If two histories produce the same available information, no policy can respond differently to them. In the sequential setting, “available information” means the relevant observational history, not merely the current sensor reading; memory can preserve an earlier distinction, but it cannot manufacture a distinction that was never sensed.

Impose a finite-state implementation and a second loss of distinctions appears. The controller no longer retains the entire observational history; it maintains a recursively updateable summary, $q_{t+1} = \delta(q_t, o_t)$. Observation identifies hidden histories that the sensors cannot distinguish, and finite memory identifies observed histories that the implementation cannot retain separately. These are related but different limitations, and they have different repairs: more memory can repair forgetting, never blindness. Moreover, not every compression of histories is implementable sequentially — the summary must be updateable from its current state and the next input alone. Causality constrains the representation itself, not only the behaviour. This is the deepest bridge between the two pillars: both ask whether behaviour remains realizable after distinctions have been identified, but they constrain different stages of the identification.

Two currencies govern everything in the corpus. Width, or rate, is the capacity of the interface: $\log|O|$, treewidth, state count, Hankel rank, determination index, Weihrauch degree — each domain names the same quantity in its own units. Distortion, or obstruction, is the deficiency of the lift: a fibre conflict, an obstruction datum, a singular value σ_k , an inner/outer gap. Every theorem in the corpus is a statement about one currency conditional on the other: either a width that suffices for a given distortion, or a distortion floor that no given width can beat. Figure 1 shows the architecture these two currencies organize.

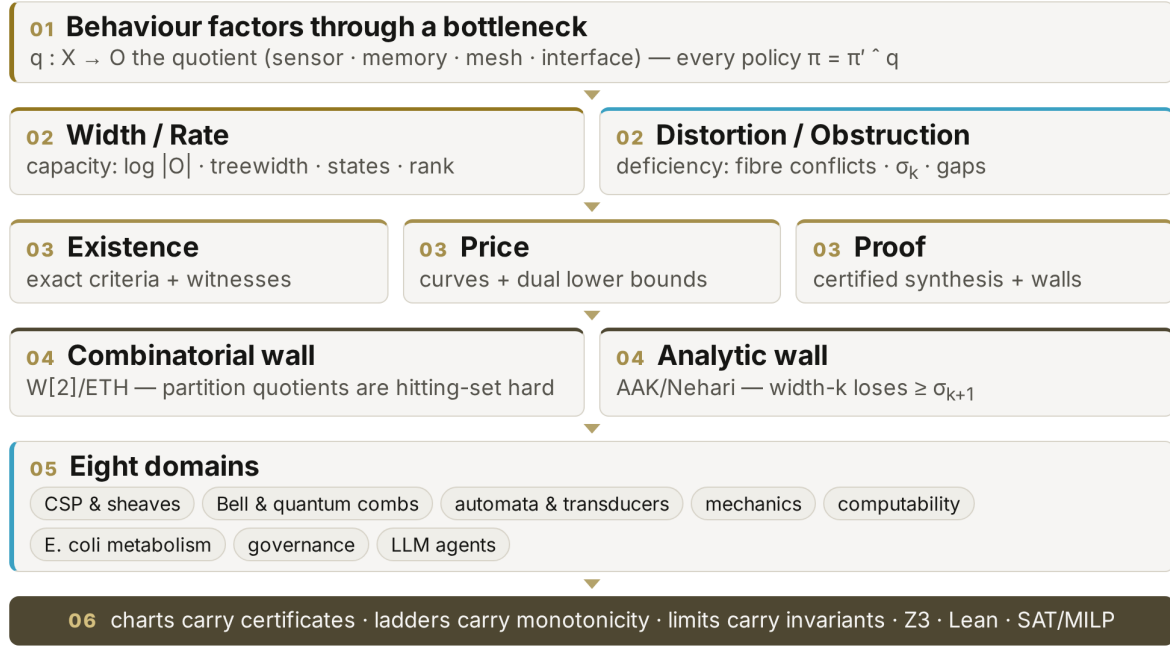


Figure 1 — The architecture of the Resolution Programme: one primitive, two currencies, three faces, two walls, eight domains, one epistemology.

The interface is the unit of cost, and the corpus measures it on two shapes. A tree-decomposition dynamic program must carry across each separator exactly enough information to glue the two sides; the minimum is governed by a gluing matrix whose rows are partial configurations on one side, columns on the other, and entries record compatibility or value. The Hankel matrix $H(u, v) = f(uv)$ is the same object on a path: the past sits on one side, the future on the other, and the present is the separator. Distinct rows give the minimal deterministic interface; the rank gives the minimal linear interface, by Fliess and Carlyle–Paz; the singular values measure what must be lost by forcing a smaller interface. Width, states, and rate are one currency — a width counted in bits of interface state is literally a rate.

3 · One Constraint, Many Disguises

Observational uniformity says that $\pi(w) = \pi(w')$ whenever the relevant sensor cannot tell w from w' ; equivalently, the policy factors through the sensor. The corpus reaches so many literatures because the same constraint recurs under different names, depending on what plays the role of the sensor. Table 1 lists the dialects now confirmed across the whole corpus, from the two described pillars to the quantum, biological, governance, and language-agent works the earlier syntheses could not resolve.

The interface acting as the “sensor”	The constraint’s name	Where it lives in the corpus
Physical sensors	Partial observability; information sets	ECD Vol 1–2; the mechanics front end
Prefix of the history	Causality	ECD causal histories; sequential transducers
The agent’s internal state	Finite memory (Myhill–Nerode)	The bounded-transduction corpus
One party’s own input	Bell locality	ECD Bell module; CHSH
Finite-precision views of the input	Continuity; Weihrauch computability	ECD computability interface
The process interface itself	No-carrying; the intercept principle	Quantum combs; environmental currying
The next-token vocabulary	Autoregressive myopia	The chess-LLM society
The aggregation grid	Composition concealment	ECOMOD biocapacity drawdown
The sampling regime	Sampled governance	Sustainability paper 5 (v47)

Table 1 — The one constraint, heard in nine dialects

Two rows are the non-obvious ones, and they carry the most weight. Bell locality: a local hidden-variable model is exactly a global policy in which each party’s output, given the hidden variable, is uniform across the other party’s setting — nonlocality is a factorization failure, which is why the CHSH inequality functions as a certificate in the same slot a Farkas vector occupies elsewhere. Computability: computable maps on infinite data are continuous — each finite piece of output is uniform across inputs that agree to some finite precision — so the “uniform” of observational uniformity and the “uniform” of Weihrauch’s uniform reducibility are the same word for the same reason. The title of the first pillar does double duty in the same way: “effective” names both effective descent in the categorical sense and effectivity in the computable sense.

The bottom rows generalize the notion of a sensor beyond hardware. An aggregation grid, a reporting convention, a next-token vocabulary, and a sampling regime are as much quotients as a photodetector: each merges worlds that a policy or an institution must then treat identically. This is why the governance corpus and the chess paper belong to the same mathematics as the control and automata works — not by analogy, but because the same factorization constraint, run through the same three questions of existence, price, and proof, produces their results.

4 • The Ladder of Assembly and the Hankel Bridge

The first pillar organizes failure as a tower of diagnostics. The obstruction datum reports the lowest failing rung and its location; each rung has its own finite witness and its own classical tool, and the classical tools — compactness, Helly, join-trees — each turn a global or infinite check into a finite local one, along three different axes: compactness along the number of worlds, Helly along the dimension of the action space, join-trees along the topology of the contexts. The second pillar adds a quantitative rung on top, and the two together form the ladder every domain in the corpus climbs:

1. Local validity — does each fibre admit one uniform admissible action? Failure witness: a fibre conflict, merged worlds with no common action; at most $d+1$ of them suffice for compact convex actions in $\mathbb{R}^d$ (Helly), finitely many for compact ones.
2. Compatibility — do local choices agree on overlaps? Failure witness: a gluing obstruction, necessarily supported on cyclic context structure; probabilistically a Farkas vector, that is, a Bell inequality. Acyclic context structure makes local consistency sufficient — the join-tree theorem, in the tradition of Vorob'ev.
3. Effective realizability — is there a computable global policy? Failure witness: a reduction showing every global policy computes, for example, a DNR function; the Weihrauch degree measures the gap. Compactness gives existence, not computability — computable trees can lack computable paths.
4. Bounded realizability — is there one within the resource bound? Failure witness for memory: $k+1$ pairwise distinguishable histories, or a Hankel block of rank $k+1$, certifying that no k -state realization exists.
5. Rate-distortion — with k states, how close can any implementation get? The floor: the $(k+1)$ -st Hankel singular value, by AAK/Nehari-type inequalities.

The top rung is why the transduction corpus is a family member rather than a neighbour. The certificates of the first pillar are the distortion-zero slice of the second pillar's curves: “no k -state machine does this exactly” is a rank statement, and “none gets within ε ” is a singular-value statement. The bridge object is the Hankel matrix. Two histories with different rows cannot share a state — a fibre conflict, in the first pillar's vocabulary; its rank is the minimal fibre count for exact realizability; and where exact realizability fails, AAK and Nehari turn the boolean obstruction (“it fails here”) into a metric one (“it fails by at least σ_{k+1} ”).

This yields the headline identification of the unified reading, due to opus 5 and confirmed by the corpus at large: the obstruction datum of effective causal descent is the qualitative Hankel spectrum of a sensor bottleneck, and the AAK singular values are the quantitative Čech theory of a memory bottleneck.

Nehari’s theorem is a descent theorem — distance-to-extension equals norm-of-obstruction — and AAK is graded obstruction theory. Once this is seen, the four presentation-relative diagnostics of the first pillar read as a degenerate, zero-or-infinity rate–distortion curve, and the multiletter AAK bounds read as the continuous completion of the same ladder. The four external syntheses reached this identification from two works alone; the quantum, channel, and instrument results surveyed in Chapter 8 supply its hardest instances.

5 • The Three Faces of Every Result

Each work in the corpus proves, for its domain, three theorems of a fixed type. First, an exact criterion with obstruction: the lift exists if and only if the obstruction vanishes, and otherwise a canonical datum locates the first failure. Second, a frontier with a dual lower bound: a Pareto curve of rate against distortion, with the lower bound supplied by duality — never by construction. Third, constructive synthesis with a hardness wall: a width-parameterized algorithm attaining the frontier, plus a proof that the parameterization is optimal, so that no cheaper route exists.

The instances cover every domain. The causal-descent pair contributes the diagnostics and obstruction datum, the Bell repair profiles, and the certifying dynamic program over tree decompositions, with NP-completeness, $W[2]$ -hardness, and ETH-tightness as its wall. The transduction manuscript contributes the Myhill–Nerode and Hankel-rank criteria, the AAK/Nehari-type converses, and the Schmidt-pair truncation ladders — each step certified by the next singular value. The combs paper contributes the intercept principle as its exact criterion and affine non-representability as its obstruction; the channels paper contributes exact diamond-ball geometry, antipodal widths as dual lower bounds in the tradition of Borsuk–Ulam, and query-uniform compression bounds as its frontier; the instruments paper completes the flat-width classification — the exact criterion for the whole affine interface. The circuits manuscript contributes the statistical face — Born-record large deviations — with Z_3 exact closure as its certified computation and a PostBQP result as its wall. The metabolic pair contributes an empirical frontier, and the governance corpus its minimax dual certificates and forecast ladders.

The trichotomy is why the corpus never confuses three different refusals: “no exact lift” (an obstruction), “no cheaper lift” (a frontier), and “no cheaper search for the lift” (a hardness wall). The five-way obstruction taxonomy of the astra synthesis — insufficient information, incompatible local obligations, ineffectivity, insufficient implementation resources, irreducible approximation error — is the diagnostic version of the same discipline, and it is actionable: a failure caused by missing information will not disappear through better optimization, a failure caused by insufficient memory does not require new sensors, and an inconclusive numerical abstraction is not a proof that the underlying mechanical specification is impossible. The governing principle is uniform across the corpus: diagnose the obstruction before choosing the repair.

6 • The Two Walls and the Price Schedule

Reading the corpora together exposes a structural fact invisible from inside any one of them: the feasible use of a bottleneck is cheap; choosing the bottleneck is hard; and even the best choice has an analytic floor. Hence two independent lower-bound technologies, which the programme shows to be the two walls of the same room. The combinatorial wall: sensor design and sensor repair are $W[2]$ -hard, admit no better than logarithmic approximation unless $P = NP$, and are ETH-tight — the signature of hitting set. The analytic wall: AAK/Hankel floors — even the right quotient of a given width cannot beat σ_k . Neither wall is visible from inside a single work; together they say that partial observability is not a nuisance to be engineered away but a conserved quantity with a price schedule.

The reason for the division of labour is precise, and it belongs to fable's synthesis: a partition quotient is a 0/1 projection, and a subspace quotient is an orthogonal projection; relaxing the former to the latter converts hitting-set into an eigenvalue problem. The integrality gap between partition and subspace quotients is therefore the price of physical realizability. The practical consequence is a division of labour that runs through the whole corpus: choosing the sensor is provably hard even to approximate, while pricing it is spectral and computable in time polynomial in the size of a finite section — the converses do not solve design, but they bound it from below cheaply.

Bottleneck	The measured price	Certificate
Absent-minded driver, 0 bits	Deterministic value 1; behavioural 4/3 (maximum of $4p - 3p^2$)	Two-world fibre conflict; a single landmark is a size-1 repair
Absent-minded driver, 1 bit	Value 4	Achievability: split the fibre
CHSH, 0 bits of communication	Win probability at most 3/4	CHSH inequality as a Farkas vector
CHSH, 1 bit	Win probability 1	One-bit protocol
Quantum instruments	In-radius $2/(n d_B \min\{d_A, d_B\})$; qutrit flat-width exactly 4/3	Complete flat-width classification
Uniform mass-spring chain	$(K^{-1})_{ij} = \min(i,j)/k$: no decay, maximal combinatorial width; every past-future block rank 1	Rank identity; two scalars per cut
Monitored circuits, $L = 8/12/16$	$\Lambda(2) = -0.1219 / -0.1239 / -0.1245$ (exact)	Z3 two-sided certificates
Born-record ESS exponent	0.0340 / 0.0349 / 0.0353 (corrected)	Exact-arithmetic closure suite
E. coli transcription layer	$r = +0.395$ (424 genes); protein-layer null $r = -0.083$	Empirical differential test
Viability fixed point	Contraction constant 0.697; fatigue exponent 3/2	Seven-optic Banach iteration

Table 2 — The price schedule: measured prices and their certificates

The schedule deserves a reading, because it is the corpus’s empirical identity card. The absent-minded driver and CHSH are the same table in two dialects — zero bits against one bit, $4/3$ against 4, $3/4$ against 1 — and each entry carries its certificate, which is what makes the price a theorem rather than an observation. The physics entries measure the same currency in continuum mechanics: the uniform chain’s inverse $(K^{-1})_{ij} = \min(i,j)/k$ decays not at all, so thresholding fails and combinatorial width is maximal, yet every past–future block has rank 1 and each cut needs only two scalars — total load and first moment — which is why rank-aware methods stay polynomial where width explodes. The quantum entries are exact rather than asymptotic: the instruments geometry closes the integrality gap by hand, and the monitored-circuit entries close it by machine, with Z3 two-sided certificates. Glover’s Hankel-norm model reduction is the analytic shadow of sensor repair, and balanced truncation the analytic shadow of choosing which states to observe.

7 • The Corpus Map: All Works on One Grid

The four external syntheses each ended with a slot-grid: a matrix of domain against face, the two described works placed, and empty cells marked with the notation of an unfinished proof. Because the remaining works have now been read in full, the grid can be closed. Table 3 docks every corpus line — the two pillars, the quantum trio, the channel and instrument papers, the monitored-circuit manuscript, the metabolic pair, the sustainability arena, and the chess paper — onto the bottleneck it studies, the slot it fills, and the signature results it contributes.

Corpus line	The bottleneck	Slot in the programme	Signature results
Effective Causal Descent, Vol 1 (v47)	Sensor quotient: worlds to observations	Qualitative core: the constraint and its disguises; diagnostics; morphisms	Obstruction datum; selector kernel; prediction games; Bell module; Weihrauch/DNR interface
ECD, Vol 2 (v14)	The same, plus the mechanics front end	Algorithms, certificates, design complexity	Width-based certifying DP; sensor repair (W[2]-hard, ETH-tight); inner/outer sandwich; exact rational influence rows
Bounded transduction (v9; 239 pp, 207 results)	Memory quotient: histories to Nerode classes	Quantitative core: the frontier	Hankel rank criteria; AAK/Nehari-type converses; determination-index calculus; multiletter AAK (conditional)
Quantum combs (submission 2)	The process interface	Quantum face of the constraint	Intercept principle; normalization-defect calculus; no right or left adjoint; affine non-representability
Environmental currying	Environment tensoring	Categorical bridge	No right adjoint except the trivial environment; recovers the no-programming theorem

Corpus line	The bottleneck	Slot in the programme	Signature results
No universal process currying	Process currying	Obstruction bridge	The intercept principle as a universal obstruction
Quantum channels (v15; 46 pp)	Query access to channels	Quantum price face	Exact diamond balls; antipodal widths (Borsuk–Ulam); query-uniform compression bounds
Quantum instruments (v18; 53 pp)	The instrument affine hull	Quantum exact geometry	Affine dimension $d_A^2(nd_B^2-1)$; in-radius; complete flat-width classification; qutrit 4/3
Monitored circuits (v22 + supplement v5)	The measurement record	Statistical estimation of the frontier	Exact replica mechanics; Born-record large deviations; Z3 closure; PostBQP result
Metabolic curvature (v22)	The FBA constraint switch	Empirical landing	κ_μ measure; 93.4–100% second-order response; $r = +0.395$ against a -0.083 null
Viability companion (v15)	The viability sheaf	Compositional home; empirical	SAVGS; StCon(B) gluing; small-loop viability-holonomy; seven-optic contraction 0.697; terminal-coalgebra maxRAF; HoTT extension
General sustainability (Round 32)	Assessment, sampling, aggregation	Governance scale	Obstruction calculus; material ledgers; sampled governance; forecast ladders; ECOMOD identifiability
Chess-LLM society	The next-token interface	AI application	Six-role society; CBS-DL; CSR theorem; amplification bound

Table 3 — The corpus map: every work docked into the programme

Four patterns emerge from the filled grid, and they are the substance of the unification. First, the quantitative face of contextuality — the cell opus 5 guessed would be filled by “rate–distortion of nonlocality” — is filled twice over: by query-uniform compression bounds for channels and by the exact affine geometry of instruments, so the quantum interfaces carry their own price lists, in closed form. Second, certified synthesis on the sequential side — the algorithmic companion the automata line lacks — remains the genuinely open cell: the spectral converses bound it, but nothing yet constructs it. Third, the statistical estimation of the frontier, guessed empty, is filled by Born-record large deviations and by the finite-sample Bell tests of the first pillar — the record of a monitored circuit is itself a bounded observer, and its large-deviation rate function estimates the frontier from data. Fourth, the trust layer is filled with named artifacts: Lean’s InterceptCert, the Z3 closure suites, the pinned reproducible computational studies.

The map also settles a doubt the four syntheses could not: the trailing DOIs and identifiers that arrived unread are companion deposits of this same programme — texts, artifacts, and datasets behind the pillars — not foreign works requiring the picture to be revised. Nothing in the corpus adds a rung or contradicts the architecture; everything fills a slot the architecture predicted. That is the strongest evidence the unification is real rather than projective: the grid was drawn from two works, and the other eleven docked without forcing.

8 • The Quantum Wing

In the quantum works the quotient is a process interface — a comb, an environment tensoring, a query, an instrument — and the constraint acquires its sharpest statements. The combs manuscript states it as a normalization defect and an intercept principle: what a comb can realize is bounded by what its interface intercepts, and the affine non-representability formula $d_C^2(d_A^2-1) + d_C^2d_A^2d_B^2(d_D^2-1)$ counts exactly which points of the ambient affine space no physically realizable comb reaches. The adjoint results are the categorical form of the same verdict: no right adjoint and no left adjoint — the interface cannot be transcended by pure algebra. Environmental currying sharpens this further: tensoring by the environment admits a right adjoint only for the trivial environment, and the failure recovers the no-programming theorem. No universal process currying then isolates the intercept principle itself as the universal obstruction — the quantum statement that not every global behaviour factors through a chosen process interface.

The channel and instrument papers are the quantum price face, and they give the frontier its exact geometry. Query-uniform compression bounds are the quantum rate–distortion statement: how much any uniform query protocol must lose when it compresses channel discrimination to a bounded query budget. The exact diamond balls and antipodal widths of the channel paper make the dual lower bound geometric — the Borsuk–Ulam theorem supplies the antipodal bound, a topological Farkas certificate in the same slot the CHSH inequality occupies classically. The instruments paper then completes the affine interface’s classification: affine dimension $d_A^2(nd_B^2-1)$, in-radius $2/(nd_B \min\{d_A, d_B\})$, a complete flat-width classification, and the qutrit value exactly $4/3$. These are exact prices rather than asymptotics — the analytic wall with the integrality gap closed by hand, dimension by dimension.

The monitored-circuits manuscript is the statistical face of the programme at its current frontier. Exact replica mechanics and Born-record large deviations treat the measurement record of a monitored circuit as itself a bounded observer of the quantum state, and characterize the large-deviation rate function of what that observer can infer — the finite-sample estimation of the frontier, run on the hardest interface the corpus studies. The v22 exact-closure suite shows the certificate discipline operating there in earnest: Z3-verified two-sided certificates, $\Lambda(2)$ values pinned at -0.1219 , -0.1239 , and -0.1245 for $L = 8, 12$, and 16 , the ESS exponents corrected to 0.0340 , 0.0349 , and 0.0353 , and the complexity-theoretic wall stated in PostBQP terms. At this writing the v23 computation of the $L = 8$ rung is still running, at chunk 4161 of 8295 — the programme is not only a description of frontiers; it is, at this moment, computing one.

9 • The Empirical Landing: Biology as a Working Instance

The metabolic pair is where the programme touches data, and it is the corpus's only empirical landing. The main manuscript defines a discrete curvature measure κ_μ for flux-balance analysis and shows that it predicts transcriptional regulation in *E. coli*: second-order response concentrates at constraint-switch boundaries at rates between 93.4 and 100 percent; the correlation with transcriptional regulation is $r = +0.395$ across 424 genes, while the protein layer shows a null of $r = -0.083$; 66 percent of regulatory cycles show non-reversion with slope 1.00; and the cycle-response association is $\rho_S = 0.865$. The theorem map — the value-flux coupling identity, the refinement-resolution bridge, Monge-Ampère atoms — is the exact-criterion face of the programme, and the Keio and iJO1366 validation chains apply the certificate discipline to wet-lab data, strain by strain.

+0.395 transcription-layer correlation, 424 genes	−0.083 protein-layer null — the differential test	0.865 cycle-response association ρ_S
---	---	---

The differential pattern is the point, and it is characteristic of the whole programme: curvature predicts the transcription layer and not the protein layer, so the signal concentrates exactly where the resolution quotient bites. Biology reads as a working instance of the theory — the cell is a bounded observer whose regulatory policy factors through metabolic viability, and regulation concentrates at constraint-switch boundaries, where the quotient's curvature is high. The empirical claim, made quietly, is large: this is not only normative mathematics; a four-billion-year-old optimizer implements its predictions, and implements them in the layer where the theory says the obstruction lives.

The viability companion gives the same landing its compositional home. SAVGS organizes the viability geometry; StCon(B) makes the compatibility rung functorial as a sheaf of constrained behaviours; the small-loop viability-holonomy theorem prices regulation around metabolic cycles; and the seven-optic composition carries a Banach contraction with constant 0.697 — below 1, hence a constructive fixed point — with a fatigue exponent of $3/2$. The terminal-coalgebra characterization of maxRAF places autocatalytic closure at a universal fixed point of the viability interface, and the stratified homotopy-type-theoretic extension (proof-sketch status, honestly flagged as such in the manuscript itself) is the programme's own discipline applied to its own frontier. The Keio E12–E26 verdict chain closes the loop from sheaf to strain.

10 • The Governance Scale

The general-sustainability corpus — Round 32 of the arena: five core papers, the four-paper wave E, a five-layer successor manuscript, six computational notes, and the ECOMOD study from the second agent — is the programme at planetary scale, and it names itself accordingly: its second paper is an obstruction calculus. The objects are assessment systems, material ledgers, delay dynamics, sampled governance, and forecast ladders with interventions; the notes supply computational certification, worked systems, exact belief computation, and probabilistic sufficiency — the certificate discipline transposed from solvers to institutions. What the governance corpus adds that the mathematics cannot is stakes and asymmetry: a mis-sampled governance grid is not a theorem failing, it is a policy succeeding on the wrong quotient, with the error surfacing a decade later in the ledger.

Each paper runs the trichotomy at institutional scale. Assessment separation (v62) separates prediction from prescription — the realizability of policy under assessment resolution, with the two failure modes priced separately. The obstruction calculus (v55) and its notes certify when institutional constraints cannot be jointly satisfied and what relaxation would cost. Material ledgers (v32) treat accounting as a quotient on physical flows, with identifiability conditions for what the books can and cannot see. Delay dynamics (v41) is guarded feedback in institutional time. Sampled governance (v47) is literally sensor design for institutions — what must be monitored, how often, at what resolution, for a target to remain attainable. Wave E's cod and Edwards forecast ladders (v59, v16) and their interventions (v23, v15) are refinement ladders carrying monotone certified forecasts — the inner/outer sandwich at the scale of fisheries and planetary boundaries — while the minimax dual-certificates note supplies the duality face, and ECOMOD (v37) shows how aggregation conceals composition: biocapacity drawdown is unidentifiable from aggregated accounts, the resolution-limited quotient made fiscal.

Read together with the rest of the corpus, the governance works are the programme's demonstration that its refusals scale. The same three questions — can this institution's policy factor through what it monitors; what does finer monitoring cost; what certificate witnesses the answer — that organize a two-node driving puzzle organize a planetary accounting system, and the answer the corpus gives at both scales is the same: certify the obstruction, price the repair, and refuse unwitnessed optimism.

11 · The AI Wing: Bounded Observers Made of Language

The chess paper — a society of specialised LLMs overcoming autoregressive myopia — is the programme instantiated on language agents, and its diagnosis is pure corpus: the next-token interface is a quotient. A model conditioned on its own output stream cannot distinguish futures that its vocabulary merges; myopia is therefore not a bug of training but a resolution constraint of the interface, exactly as a sensor suite merges worlds and a finite state set merges histories. The repair the paper builds is the repair menu of the corpus in full: the six-role society (CBS-DL) specializes sensors, since roles attend to different features of the position; it allocates memory, since roles maintain distinct state; it re-architects computation, since roles compose; and it refines the certified model through deep lookahead, accepting bounded amplification rather than unbounded promise.

The theorems sit in the canonical slots. The CSR theorem and the amplification bound of Theorem 2* are the frontier and the wall: what amplification a society of bounded observers can buy, and the bound no such society can beat — the duality face for language agents. Deep-lookout planning is the width-parameterized synthesis. The correspondence is exact enough that the paper can be read as the programme's existence proof in silicon: bounded observers, composed under a graded interface, attaining a certified frontier that no single bounded observer reaches — which is, not coincidentally, the chess society's empirical result against monolithic baselines.

12 · The Compositional Home

The categorical works are not decoration on the corpus; they are its wiring diagram. The morphism theory of the first pillar — functoriality, sequential and parallel composition, guarded feedback — and the transducer calculus of the second — cascade, product, feedback — are the same object seen from two directions: a monoidal category with trace, doubly graded by rate and distortion. Objects are interfaces; morphisms are policies and transducers; composition adds rates and accumulates distortion subadditively; guarded feedback is the trace, guarded precisely so that the accumulation stays well-founded. This is the level at which the mechanics front end, the CSP encodings, the Bell scenarios, the automata, the optics, and the combs become the same kind of thing and can be wired together, with certificates composing alongside the behaviours they certify.

Each categorical construction in the corpus fills a slot in that home. Environmental currying is the interface algebra: the intercept principle is what remains of currying when the environment is physical, and the no-programming theorem is its obstruction wall. Sheaf gluing $\text{StCon}(\mathbf{B})$ is the compatibility rung made functorial. The seven-optic composition of the viability companion is the graded wiring of constrained behaviours, and its Banach constant (0.697) is the distortion accounting of that wiring made

constructive. The terminal-coalgebra maxRAF makes autocatalytic closure a fixed point of the viability interface — the empirical landing expressed in the compositional home — and the homotopy-type-theoretic stratification extends the home upward, at proof-sketch status, in the direction where guarded fixed points become homotopy fixed points.

One quantitative law remains genuinely open in the home, and the two pillars pose it jointly: feedback. Guarded feedback is the combinatorial twin of feedback interconnection of transducers, and the quantitative law for how obstructions compose under feedback should be small-gain-shaped in Hankel norm — distortion accumulating around a loop should be governed by the product of the loop's interface norms. No work in the corpus yet proves it; every work in the corpus is arranged as if awaiting it.

13 · The Methodological Spine

The programme's epistemology is unusual, and it is stated as a feature rather than apologised for: diagnostics, widths, and certificates are presentation-relative — properties of a chart, not of the abstract problem. Complexity belongs to a problem-presentation pair, morphisms transport it, and the curves of the second pillar are the exchange rates between presentations. The frontier is the chart-invariant content. Refinement ladders — width-adaptive sparsification, multiletter blocking, mesh refinement, tree-decomposition improvement, forecast ladders — are the mechanism by which chart-relative certificates converge monotonically to invariant truth: an inner/outer sandwich, $P_{\text{inner}} \subseteq P_{\text{true}} \subseteq P_{\text{outer}}$, closing on the real policy space from both sides. The epigram that closes opus 5's synthesis is the corpus's own answer to how proof-producing computation can speak about continua and hidden states at all: charts carry certificates; ladders carry monotonicity; limits carry invariants.

The second commitment is proof-relevant negatives: every “no” in the corpus is witnessed, and there are no unwitnessed negatives anywhere in it. Fibre conflicts witness local failure; Farkas vectors and Bell inequalities witness contextuality; DNR reductions witness ineffectivity; rank $k+1$ blocks and Hankel singular values witness memory floors; antipodal widths witness channel floors; Z3 two-sided certificates witness exact closure in monitored circuits; Lean's InterceptCert witnesses the intercept principle by proof assistant. Trust accordingly moves from solvers to small checkers, and the SAT/CP-SAT/MILP studies of the second pillar are simultaneously performance comparisons and N-version differential testing, pinned by reproducible artifacts. The sandwich separates three outcomes and refuses to collapse them: a policy in the inner set certifies feasibility; an empty outer set certifies infeasibility; and an empty inner set with a nonempty outer set marks unresolved uncertainty — never silently resolved.

Two kinds of approximation are kept rigorously distinct, and confusing them is the corpus's definition of error. Behavioural approximation deliberately tolerates distortion in the implemented behaviour; certified abstraction approximates the analysis while preserving sound conclusions about a hard specification. A small average distortion is not a worst-case safety guarantee. And when the input is a frequency table rather than an exact model, a certificate becomes a test with a controlled error rate — the finite-sample

Bell tests of the first pillar and the Born-record large deviations of the circuits manuscript are the same move, executed on classical count data and on quantum measurement records respectively.

14 • The Open Frontier

A unified picture earns its keep by making the open problems legible as cells of the same grid. The frontier of the programme, as it stands at this writing, is the following:

1. The multiletter AAK theorem — explicitly conditional in the transduction manuscript, and the deepest single open problem in the quantitative core. The block-length ladder is the continuous completion of the assembly ladder; its missing rung is the analytic wall's top course.
2. Certified synthesis on the sequential side — the one slot the filled grid still leaves empty: the algorithmic companion for automata and transducers that the certifying dynamic program is for contexts. The spectral converses bound it; nothing yet constructs it.
3. The partition-subspace bridge — a theorem, not a vocabulary, equating the integrality gap between 0/1 and orthogonal projections with the price of physical realizability, with spectral certificates (Courant–Fischer witnesses, checkable by exact semidefinite tests) joining the typed certificate family.
4. The graded category with trace, and the small-gain law in Hankel norm for how distortion composes around guarded feedback — the compositional home's missing quantitative law.
5. Rank-aware width for physics — dynamic programs carrying sufficient statistics (total load and first moment per cut) that stay polynomial where combinatorial width explodes, with Hankel spectra as their natural converse; the rank-1 blocks of the chain inverse are the paradigm.
6. Statistical estimation of obstruction data — Born-record large deviations as the beginning of a quantitative obstruction profile: a distortion floor for each rung of the ladder, estimated from records rather than computed from models.
7. The three named open problems of the transduction line — intrinsic multi-alphabet Hankel equality, the exact symbolic grounding gap, and the non-degenerate two-term active decomposition.
8. Live computations — the v23 $L = 8$ rung (at chunk 4161 of 8295 at this writing), the figshare DOI, and the union citation list: the corpus's own pre-submission blockers, tracked with the same status discipline as its theorems.

For a reader entering the corpus with this picture in hand, the reading order that follows the architecture rather than the chronology is: Effective Causal Descent, Volume 1, then Volume 2; the unified transduction manuscript (v9); the combs submission with the two currying papers; the channel and instrument papers; the monitored-circuits manuscript with its supplement; the metabolic pair; the sustainability compendium of Round 32; and the chess paper — with the master file, “top-down of my work.txt”, as the prism through which the whole is meant to refract.

Read whole, the corpus is a single instrument played across eight registers, and its coherence is architectural rather than thematic: one primitive, two currencies, three faces, two walls, one compositional home, one epistemology. What the union asserts that no single volume does is exactly what opus 5 said, now confirmed across every line of the corpus: contextuality, finite-state approximation, and treewidth-bounded synthesis are one subject — bounded-interface factorization; Farkas, Nehari, Čech, and dynamic-programming refutation certificates are one duality, instantiated four ways; every “no” is proof-relevant; hardness comes in exactly two flavours, complementary rather than competing; and presentation-relativity plus monotone refinement is a sufficient epistemology for certified reasoning about partially observed physical systems. That is the grand picture. The works are its witnesses.